

THE UNIVERSITY OF AZAD JAMMU & KASHMIR
MUZAFFARABAD

DATABASE MANAGEMENT SYSTEM LAB

Lab 6 — SQL Filters — Comparison and Logical Operators
Employee & Bookstore Databases

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Roll No	2024-SE-11
Semester	04
Course Title	Database Management System (CS-2204)
Submitted To	Engr. Awais Rathore
Lab No.	06
Tool Used	XAMPP (MySQL / MariaDB) with phpMyAdmin

1. Objective

- Use the WHERE clause with comparison operators (=, <, >, <=, >=, <>).
- Combine conditions using logical operators AND, OR, and NOT.
- Use BETWEEN, IN, LIKE, and IS NULL for pattern and range-based filtering.
- Sort and limit results using ORDER BY and LIMIT.

2. Database Structure

Two databases are used: filters_lab (an Employee table) for guided practice, and bookstore_lab (a Book table) for a standalone graded assessment.

2.1 filters_lab

Table	Purpose
Employee	Employee records used to practice comparison and logical filtering

2.2 bookstore_lab

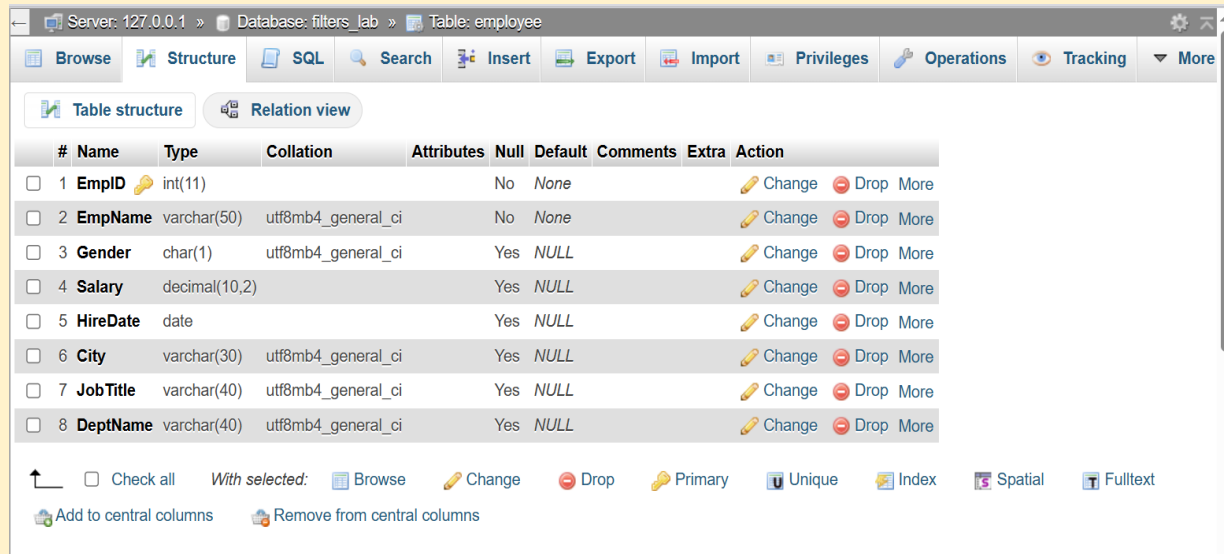
Table	Purpose
Book	Book catalog used for the graded filtering assessment

3. Key Tasks / Concepts Implemented

- Filtering employees by salary range, city, and gender using comparison and logical operators.
- Using BETWEEN and IN to filter on ranges and value lists.
- Using LIKE with wildcards for pattern-based name searches.
- Using IS NULL / IS NOT NULL to find missing data.
- Sorting results with ORDER BY and restricting result size with LIMIT.
- Applying the same filtering techniques to the Book table for the assessment.

4. Implementation (phpMyAdmin)

All filter queries were written and executed against both databases using phpMyAdmin under XAMPP.



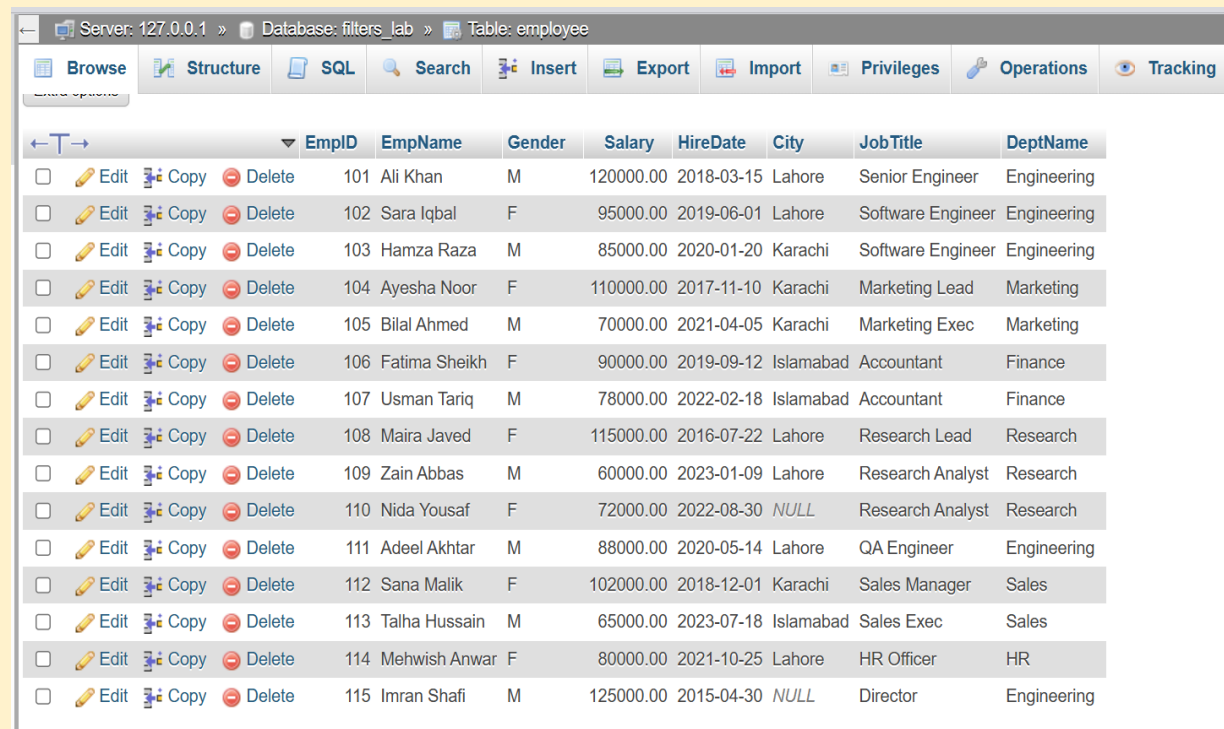
Server: 127.0.0.1 » Database: filters_lab » Table: employee

Table structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 EmpID	int(11)			No	None			
☐	2 EmpName	varchar(50)	utf8mb4_general_ci		No	None			
☐	3 Gender	char(1)	utf8mb4_general_ci		Yes	NULL			
☐	4 Salary	decimal(10,2)			Yes	NULL			
☐	5 HireDate	date			Yes	NULL			
☐	6 City	varchar(30)	utf8mb4_general_ci		Yes	NULL			
☐	7 JobTitle	varchar(40)	utf8mb4_general_ci		Yes	NULL			
☐	8 DeptName	varchar(40)	utf8mb4_general_ci		Yes	NULL			

☐ Check all With selected: Browse Change Drop Primary Unique Index Spatial Fulltext

Add to central columns Remove from central columns



Server: 127.0.0.1 » Database: filters_lab » Table: employee

Browser

	EmpID	EmpName	Gender	Salary	HireDate	City	JobTitle	DeptName
☐	101	Ali Khan	M	120000.00	2018-03-15	Lahore	Senior Engineer	Engineering
☐	102	Sara Iqbal	F	95000.00	2019-06-01	Lahore	Software Engineer	Engineering
☐	103	Hamza Raza	M	85000.00	2020-01-20	Karachi	Software Engineer	Engineering
☐	104	Ayesha Noor	F	110000.00	2017-11-10	Karachi	Marketing Lead	Marketing
☐	105	Bilal Ahmed	M	70000.00	2021-04-05	Karachi	Marketing Exec	Marketing
☐	106	Fatima Sheikh	F	90000.00	2019-09-12	Islamabad	Accountant	Finance
☐	107	Usman Tariq	M	78000.00	2022-02-18	Islamabad	Accountant	Finance
☐	108	Maira Javed	F	115000.00	2016-07-22	Lahore	Research Lead	Research
☐	109	Zain Abbas	M	60000.00	2023-01-09	Lahore	Research Analyst	Research
☐	110	Nida Yousaf	F	72000.00	2022-08-30	NULL	Research Analyst	Research
☐	111	Adeel Akhtar	M	88000.00	2020-05-14	Lahore	QA Engineer	Engineering
☐	112	Sana Malik	F	102000.00	2018-12-01	Karachi	Sales Manager	Sales
☐	113	Talha Hussain	M	65000.00	2023-07-18	Islamabad	Sales Exec	Sales
☐	114	Mehwish Anwar	F	80000.00	2021-10-25	Lahore	HR Officer	HR
☐	115	Imran Shafi	M	125000.00	2015-04-30	NULL	Director	Engineering

Structure and browser tab for Employee table

Server: 127.0.0.1 » Database: filters_lab » Table: Employee

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking

Showing rows 0 - 5 (6 total, Query took 0.0011 seconds.)

```
SELECT EmpID, EmpName, Salary FROM Employee WHERE Salary > 90000;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	EmpID	EmpName	Salary
☐ Edit Copy Delete	101	Ali Khan	120000.00
☐ Edit Copy Delete	102	Sara Iqbal	95000.00
☐ Edit Copy Delete	104	Ayesha Noor	110000.00
☐ Edit Copy Delete	108	Maira Javed	115000.00
☐ Edit Copy Delete	112	Sana Malik	102000.00
☐ Edit Copy Delete	115	Imran Shafi	125000.00

Check all | With selected: Edit Copy Delete Export

Server: 127.0.0.1 » Database: filters_lab » Table: Employee

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

Showing rows 0 - 5 (6 total, Query took 0.0003 seconds.)

```
SELECT * FROM Employee WHERE Gender = 'F' AND DeptName != 'Engineering';
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	EmpID	EmpName	Gender	Salary	HireDate	City	JobTitle	DeptName
☐ Edit Copy Delete	104	Ayesha Noor	F	110000.00	2017-11-10	Karachi	Marketing Lead	Marketing
☐ Edit Copy Delete	106	Fatima Sheikh	F	90000.00	2019-09-12	Islamabad	Accountant	Finance
☐ Edit Copy Delete	108	Maira Javed	F	115000.00	2016-07-22	Lahore	Research Lead	Research
☐ Edit Copy Delete	110	Nida Yousaf	F	72000.00	2022-08-30	NULL	Research Analyst	Research
☐ Edit Copy Delete	112	Sana Malik	F	102000.00	2018-12-01	Karachi	Sales Manager	Sales
☐ Edit Copy Delete	114	Mehwish Anwar	F	80000.00	2021-10-25	Lahore	HR Officer	HR

Check all | With selected: Edit Copy Delete Export

Examples of Comparison/Logical operators used

Server: 127.0.0.1 » Database: filters_lab » Table: Employee

Showing rows 0 - 1 (2 total, Query took 0.0002 seconds.)

```
SELECT EmpName FROM Employee WHERE EmpName LIKE 'M%';
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	EmpName
☐ Edit Copy Delete	Maira Javed
☐ Edit Copy Delete	Mehwish Anwar

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Server: 127.0.0.1 » Database: filters_lab » Table: Employee

Showing rows 0 - 5 (6 total, Query took 0.0002 seconds.)

```
SELECT * FROM Employee WHERE HireDate BETWEEN '2020-01-01' AND '2022-12-31';
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	EmpID	EmpName	Gender	Salary	HireDate	City	JobTitle	DeptName
☐ Edit Copy Delete	103	Hamza Raza	M	85000.00	2020-01-20	Karachi	Software Engineer	Engineering
☐ Edit Copy Delete	105	Bilal Ahmed	M	70000.00	2021-04-05	Karachi	Marketing Exec	Marketing
☐ Edit Copy Delete	107	Usman Tariq	M	78000.00	2022-02-18	Islamabad	Accountant	Finance
☐ Edit Copy Delete	110	Nida Yousaf	F	72000.00	2022-08-30	NULL	Research Analyst	Research
☐ Edit Copy Delete	111	Adeel Akhtar	M	88000.00	2020-05-14	Lahore	QA Engineer	Engineering
☐ Edit Copy Delete	114	Mehwish Anwar	F	80000.00	2021-10-25	Lahore	HR Officer	HR

☐ Check all | With selected: Edit Copy Delete Export

Examples of LIKE, BETWEEN, IN queries

5. Conclusion

This lab built practical fluency in filtering relational data with comparison and logical operators, pattern matching, and result ordering — core tools used in almost every real-world query.